Aarhus University

Department Of Nerds (Math)

Introduction to Algebra 1

Into The Omegaverse

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Goofy ah Math is scawy oWo

Exercise: In the following, n denotes a natural number.

1. Show, by induction on n, that
$$\sum_{k=1}^{n} \frac{1}{k^2} \leq 2 - \frac{1}{n}.$$
Don't worry Pwincesess, your Alpha is here...
The base case for $n = 1$
$$\text{LHS:} \quad \frac{1}{1^2} = 1 \qquad \text{RHS:} \quad 2 - \frac{1}{1} = 1$$
the base case is satisfied since $1 \leq 1$
Next step is the hypothesis for induction ThingyMaBob, which is to assume it holds for some positive number Ω = n (Queue introduction to the Omegaverse) #FanficForLife
The Goofy ah Ω goes to town #Purr
$$\sum_{k=1}^{\Omega} \frac{1}{k^2} \leq 2 - \frac{1}{\Omega}$$
Now it is time for marriage, which is to prove it for $\Omega + 1$, the next number in the chain, so the actual induction step.
But remember as a true Alpha, we know the left side is just the summation:
$$\frac{1}{1^2} + \frac{1}{2^2} + \ldots + \frac{1}{\Omega^2}$$
This means the next one in the series is just $\frac{1}{(\Omega+1)^2}$ so the summation for $\Omega+1$ becomes
$$\sum_{k=1}^{\Omega+1} \frac{1}{k^2}$$
And for the right hand side, the next number is just added as so #HowlingForTheMoon
$$\sum_{k=1}^{\Omega+1} \frac{1}{(k^2} = 2 - \frac{1}{\Omega} + \frac{1}{(\Omega+1)^2}$$

From here it's a matter of transformation (under the fullmoon #WerewolfBaddie)

$$2 - \frac{1}{\Omega} + \frac{1}{(\Omega+1)^2} \tag{1}$$

$$2 - \frac{1 \cdot (\Omega+1)^2 + \Omega \cdot 1}{\Omega(\Omega+1)^2} \tag{2}$$

$$2 - \frac{(\Omega+1)^2 + \Omega}{\Omega(\Omega+1)^2} \tag{3}$$

$$2 - \frac{\Omega^2 + 2\Omega + 1 + \Omega}{\Omega(\Omega+1)^2} \tag{4}$$

$$2 - \frac{\Omega^2 + \Omega}{\Omega(\Omega+1)^2} - \frac{2\Omega+1}{\Omega(\Omega+1)^2} \tag{5}$$

$$2 - \frac{\Omega(\Omega+1)}{\Omega(\Omega+1)^2} - \frac{2\Omega+1}{\Omega(\Omega+1)^2} \tag{6}$$

$$2 - \frac{1}{\Omega+1} - \frac{2\Omega+1}{\Omega(\Omega+1)^2} \tag{7}$$

2. Conclude that for all natural numbers n, it holds that

$$\sum_{k=1}^{n} \frac{1}{k^2} < 2.$$

3. Consider whether the conclusion in 2 can be shown directly by induction on n.